Key Differences

Feature	**Ethernet Cables (Copper/Twisted Pair)**	**Fiber Optic Cables**
Transmission Medium	Transmits data using **electrical signals** over copper wires.	Transmits data using **light pulses** through glass or plastic strands.
Distance & Signal Loss	Limited to short distances (maximum **100 meters** or 328 feet) before the signal degrades.	Can travel **tens of kilometers** without significant signal degradation or needing a repeater.
Interference Resistance	Vulnerable to **Electromagnetic Interference (EMI)** from power lines, fluorescent lights, and motors.	Completely **immune to EMI** and radio frequency interference since light is non-conductive.

Real-World Applications

1. Where Ethernet is Used: Home LAN/Wi-Fi Router Connections

Ethernet cables (like Cat6) are perfect for connecting a local smart TV, gaming console, or desktop PC directly to a **home Wi-Fi router**. Because the distances inside a house are well under 100 meters, copper is the most cost-effective and flexible choice for reliable, low-latency local connections.

2. Where Fiber Optics are Used: College Campus Backbone

A **college campus backbone** relies heavily on fiber optic cables to interconnect separate academic buildings, libraries, and dormitories spread across wide physical areas. Fiber easily spans the long distances between buildings and can be run underground alongside high-voltage power grids without suffering from electrical interference.